

Construction of Mutually Unbiased Bases (MUB)

1 Introduction

A d -level quantum system lives inside a d -dimensional Hilbert space ($\mathcal{H}$). The most general way to describe a quantum state is using a $d \times d$ density matrix ρ in $\mathcal{H}$, which is positive semi-definite, Hermitian, and has a unit trace. This gives us $d^2 - 1$ degrees of freedom (independent real parameters) for the complete determination of a quantum state. An orthogonal quantum measurement performed on such a system gives us $d - 1$ independent probabilities. Hence, we need $(d + 1)$ different independent orthogonal measurements to completely determine a quantum state without any redundancy or loss of information. The set of measurements will not have any possible overlaps if they mutually differ as much as possible, so that each of them provides new information about the quantum state.

It has been shown that measurement in Mutually Unbiased Bases (MUB) provides the optimal way to determine a quantum state. If the outcome of a measurement in one MUB is known with certainty, then all outcomes of a measurement in any unbiased basis are equally likely, because measurements in mutually unbiased bases are maximally incompatible.

Notations

We define $\mathcal{H} \cong \mathbb{C}^d$ to be a d -dimensional complex Hilbert space. Throughout the exposition, we assume d to be finite. For two unit vectors $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, the scalar product is defined by $\langle\phi|\psi\rangle$.

The space of linear operators on $\mathcal{H}$ is denoted $\mathcal{B}(\mathcal{H})$, which is a d^2 -dimensional complex vector space. For $X, Y \in \mathcal{B}(\mathcal{H})$, the scalar product is defined by

$$\langle X, Y \rangle = \text{Tr}(X^\dagger Y).$$

- A *quantum state* is represented by an operator in $\mathcal{B}(\mathcal{H})$ which is Hermitian, positive-semidefinite, and has unit trace.
- *Orthogonal projection operators* $P \in \mathcal{B}(\mathcal{H})$ are those operators which satisfy

$$P^2 = P, \quad P^\dagger = P.$$

A one-dimensional projection operator can be denoted by $|\psi\rangle\langle\psi|$, for a unit vector $|\psi\rangle \in \mathcal{H}$.

- An *elementary measurement* $\mathcal{M} = \{P_0, P_1, \dots, P_{d-1}\}$ is a family of d mutually orthogonal one-dimensional projection operators P_j , $j = 0, 1, \dots, d - 1$, such that

$$\sum_{j=0}^{d-1} P_j = I.$$

If the measurement $\mathcal{M}$ is performed when the state of the system is say $|\phi\rangle\langle\phi|$, the result of such a measurement is one of the classical outcomes $j \in \{0, 1, 2, \dots, d-1\}$ with probability $\text{Tr}(\rho P_j) = \text{Tr}(|\phi\rangle\langle\phi| P_j) = |\langle\psi|\phi\rangle|^2 = p_j$ for each j .

For an elementary measurement $\mathcal{M}$, we associate the operator algebra

$$\mathcal{A}(\mathcal{M}) = \left\{ \sum_{j=0}^{d-1} c_j P_j : c_j \in \mathbb{C}, j = 0, 1, \dots, d-1 \right\}.$$

This is a space spanned by the projection operators of the measurement $\mathcal{M}$ (a subspace of $\mathcal{B}(\mathcal{H})$). The subalgebra $\mathbb{C}I \subset \mathcal{A}(\mathcal{M})$ consists precisely of the constant-valued operators in $\mathcal{A}(\mathcal{M})$.

Definition 1. Two elementary measurements $\mathcal{M} = \{P_0, \dots, P_{d-1}\}$, $\mathcal{N} = \{Q_0, \dots, Q_{d-1}\}$ are said to be **weakly mutually unbiased (WMUB)** if

$$\mathcal{A}(\mathcal{M}) \cap \mathcal{A}(\mathcal{N}) = \mathbb{C}I,$$

and **strongly mutually unbiased (SMUB)** if the subspaces

$$\mathcal{A}(\mathcal{M}) \ominus \mathbb{C}I \quad \text{and} \quad \mathcal{A}(\mathcal{N}) \ominus \mathbb{C}I$$

of the space $\mathcal{B}(\mathcal{H})$ are mutually orthogonal.

(For $S_1 \subseteq S_2 \subseteq \mathcal{B}(\mathcal{H})$, $S_2 \ominus S_1$ denotes the orthogonal compliment of S_1 in S_2 .)

Proposition 1. Two elementary measurements $\mathcal{M} = \{P_0, P_1, \dots, P_{d-1}\}$, $\mathcal{N} = \{Q_0, Q_1, \dots, Q_{d-1}\}$ are SMUB if and only if

$$\text{Tr}(P_i Q_j) = d^{-1} \quad \forall i, j \in \{0, 1, 2, \dots, d-1\}.$$

Proof. Two elementary measurements $\mathcal{M}$ and $\mathcal{N}$ are SMUB if $\mathcal{A}(\mathcal{M}) \ominus \mathbb{C}I$ and $\mathcal{A}(\mathcal{N}) \ominus \mathbb{C}I$ are orthogonal. So we need to find operators that span each of these two subspaces and these should be mutually orthogonal.

$\mathcal{A}(\mathcal{M}) \ominus \mathbb{C}I$ is spanned by

$$A \in \mathcal{A}(\mathcal{M}) \text{ such that } \langle I, A \rangle = 0 \iff \text{Tr}(A) = 0.$$

Take $A = P_i - \frac{1}{d}I$. Then

$$\text{Tr}(A) = \text{Tr}(P_i) - \frac{1}{d} \text{Tr}(I).$$

Since $\text{Tr}(P_i) = \text{Tr}(|e_i^{(m)}\rangle\langle e_i^{(m)}|) = \langle e_i^{(m)} | e_i^{(m)} \rangle = 1$ and $\text{Tr}(I) = d$,

$$\text{Tr}(A) = 1 - 1 = 0.$$

Hence $P_i - \frac{1}{d}I$ spans $\mathcal{A}(\mathcal{M}) \ominus \mathbb{C}I$. Similarly, $Q_j - \frac{1}{d}I$ spans $\mathcal{A}(\mathcal{N}) \ominus \mathbb{C}I$.

Now,

$$\begin{aligned} \langle P_i - \frac{1}{d}I, Q_j - \frac{1}{d}I \rangle &= 0 \\ \implies \text{Tr}[(P_i - \frac{1}{d}I)(Q_j - \frac{1}{d}I)] &= 0 \end{aligned}$$

$$\begin{aligned}
&\implies \text{Tr}[P_i Q_j] - \frac{1}{d} \text{Tr}[P_i] - \frac{1}{d} \text{Tr}[Q_j] + \frac{1}{d^2} \text{Tr}[I] = 0 \\
&\implies \text{Tr}[P_i Q_j] - \frac{1}{d} - \frac{1}{d} + \frac{1}{d} = 0 \\
&\implies \text{Tr}[P_i Q_j] = \frac{1}{d}.
\end{aligned}$$

Writing $P_i = |e_i^{(m)}\rangle\langle e_i^{(m)}|$ and $Q_j = |e_j^{(n)}\rangle\langle e_j^{(n)}|$,

$$\text{Tr}[P_i Q_j] = \text{Tr}\left[|e_i^{(m)}\rangle\langle e_i^{(m)}|e_j^{(n)}\rangle\langle e_j^{(n)}|\right] = \left|\langle e_j^{(n)}|e_i^{(m)}\rangle\right|^2.$$

Hence,

$$\left|\langle e_j^{(n)}|e_i^{(m)}\rangle\right|^2 = \frac{1}{d}.$$

□

Definition 2. Let $B_1 = \{|e_0^{(1)}\rangle, \dots, |e_{d-1}^{(1)}\rangle\}$ and $B_2 = \{|e_0^{(2)}\rangle, \dots, |e_{d-1}^{(2)}\rangle\}$ be two orthonormal bases in the d -dimensional state space. They are said to be **mutually unbiased bases (MUB)** if

$$\left|\langle e_i^{(1)}|e_j^{(2)}\rangle\right|^2 = \frac{1}{d}, \quad \text{for every } i, j = 0, \dots, d-1.$$

A set $\{B_1, \dots, B_m\}$ of orthonormal bases in $\mathbb{C}^d$ is called a **set of mutually unbiased bases (set of MUB)** if each pair B_i and B_j is mutually unbiased.

Note that the SMUB defined for one-dimensional orthogonal projection measurements is the same as standard MUB.

2 Construction of Sets of MUB for Prime Dimension

Theorem 1. Given an MUB, we can rotate by a unitary matrix $j \mapsto (j+1) \bmod d$, which gives us another MUB (which is the eigenbasis of the unitary).

Proof. Let $B_1 = \{|e_0^{(1)}\rangle, |e_1^{(1)}\rangle, \dots, |e_{d-1}^{(1)}\rangle\}$ be an orthonormal basis in $\mathbb{C}^d$. And $B_2 = \{|e_0^{(2)}\rangle, |e_1^{(2)}\rangle, \dots, |e_{d-1}^{(2)}\rangle\}$ be another orthonormal basis which is the eigenbasis of a unitary V satisfying

$$V|e_j^{(1)}\rangle = \beta_j|e_{j+1}^{(1)}\rangle, \quad |\beta_j| = 1,$$

with indices taken mod d (i.e. $|e_d^{(1)}\rangle \mapsto |e_0^{(1)}\rangle$).

We need to prove that B_1 and B_2 are MUB.

Since B_2 is the eigenbasis of V ,

$$V|e_i^{(2)}\rangle = \lambda_i|e_i^{(2)}\rangle, \quad |\lambda_i| = 1 \text{ (since } V \text{ is unitary).}$$

Now,

$$\begin{aligned}
\left|\langle e_i^{(2)}|e_1^{(1)}\rangle\right| &= \left|\langle e_i^{(2)}|V^\dagger V|e_1^{(1)}\rangle\right| \\
&= \left|\langle e_i^{(2)}|V^\dagger|e_2^{(1)}\rangle\right|
\end{aligned}$$

$$= \left| \left\langle e_i^{(2)} \middle| e_2^{(1)} \right\rangle \right|$$

Similarly,

$$\left| \left\langle e_i^{(2)} \middle| e_0^{(1)} \right\rangle \right| = \left| \left\langle e_i^{(2)} \middle| e_1^{(1)} \right\rangle \right| = \cdots = \left| \left\langle e_i^{(2)} \middle| e_{d-1}^{(1)} \right\rangle \right|.$$

Since $\sum_{j=0}^{d-1} \left| \left\langle e_i^{(2)} \middle| e_j^{(1)} \right\rangle \right|^2 = 1$ (completeness of B_1),

$$\left| \left\langle e_j^{(1)} \middle| e_i^{(2)} \right\rangle \right|^2 = \frac{1}{d}.$$

Hence B_1 and B_2 are MUB. □

Note: Throughout this section we consider d is a prime number and all algebraic operations are mod d

We now define our unitary operator V as $X_d(Z_d)^k$. Here, X_d and Z_d are a generalization of the Pauli operators on $\mathbb{C}^d$ such that

$$X_d|e_j\rangle = |e_{j+1}\rangle, \quad Z_d|e_j\rangle = \omega^j|e_j\rangle, \quad \omega = e^{2\pi i/d},$$

Theorem 2. For $0 \leq k, \ell \leq d-1$, the eigenvectors of $X_d(Z_d)^k$ are cyclically shifted under the action of $X_d(Z_d)^\ell$.

Proof. Suppose the eigenvectors of $X_d(Z_d)^k$ are $|e_i^k\rangle$ so that

$$X_d(Z_d)^k|e_i^k\rangle = \omega^j|e_i^k\rangle.$$

We may write these eigenvectors explicitly as

$$|e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{i(d-j)} \omega^{-ks_j} |e_j\rangle,$$

Here, $i = 0, \dots, d-1$ and $s_j = j + j + 1 \cdots + d-1$

Let's check this:

$$X_d(Z_d)^k|e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{i(d-j)} \omega^{-ks_j} \omega^{kj} |e_{j+1}\rangle$$

Re-indexing $j+1 \rightarrow j$

$$X_d(Z_d)^k|e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=1}^d \omega^{i(d-j+1)} \omega^{-k(s_j+j-1)} \omega^{k(j-1)} |e_j\rangle$$

Notice $s_{j-1} = s_j + j - 1$ and $j = 0$ is the same as $j = d$. So,

$$X_d(Z_d)^k|e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{i(d-j+1)} \omega^{-ks_j} |e_j\rangle = \frac{1}{\sqrt{d}} \omega^i \sum_{j=0}^{d-1} \omega^{i(d-j)} \omega^{-ks_j} |e_j\rangle = \omega^i |e_i^k\rangle$$

Now, let's see the action of $X_d(Z_d)^\ell$ on $|e_j^k\rangle$

$$X_d(Z_d)^\ell |e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{i(d-j)} \omega^{-ks_j} \omega^{\ell j} |e_{j+1}\rangle$$

Re-indexing $j+1 \rightarrow j$

$$X_d(Z_d)^k |e_i^k\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{i(d-j+1)} \omega^{-k(s_j+j-1)} \omega^{\ell(j-1)} |e_j\rangle = \omega^{i+k+l} |e_{i+k+l}^k\rangle$$

□

Also, $Z_d |e_j\rangle = \omega^j |e_j\rangle$ So,

$$\left| \langle e_j | e_i^{(k)} \rangle \right|^2 = \frac{1}{d}.$$

Hence, the set of bases each consisting of the eigenvectors of

$$Z_d, \quad X_d, \quad X_d Z_d, \quad X_d(Z_d)^2, \quad \dots, \quad X_d(Z_d)^{d-1}$$

forms a set of $d+1$ MUB.

Example

1. $d = 2$

The unitary operators whose eigenvectors form MUB are $Z_2, X_2, X_2 Z_2$. So, The set of MUBs are:

$$\begin{aligned} & \{|0\rangle, |1\rangle\} \\ & \left\{ \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \frac{|0\rangle - |1\rangle}{\sqrt{2}} \right\} \\ & \left\{ \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \frac{|0\rangle - i|1\rangle}{\sqrt{2}} \right\} \end{aligned}$$

2. $d = 3$

The set of MUBs are formed by the eigenvectors of matrices $Z_3, X_3, X_3 Z_3, X_3(Z_3)^2$.

$$Z_3|0\rangle = \omega^0|0\rangle, Z_3|1\rangle = \omega|1\rangle, Z_3|2\rangle = \omega^2|2\rangle; \omega = \exp(2\pi i/3)$$

$$X_3|0\rangle = |1\rangle, X_3|1\rangle = |2\rangle, X_3|2\rangle = |0\rangle$$

So,

$$Z_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, X_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, X_3 Z_3 = \begin{pmatrix} 0 & 0 & \omega^2 \\ 1 & 0 & 0 \\ 0 & \omega & 0 \end{pmatrix}, X_3(Z_3)^2 = \begin{pmatrix} 0 & 0 & \omega \\ 1 & 0 & 0 \\ 0 & \omega^2 & 0 \end{pmatrix}$$

2.1 Maximal Commuting Classes and Their Relation to MUB

Here, we describe the relation between maximal commuting classes of orthogonal unitary operators and MUB.

The main idea is: one maximal commuting class will have one common eigenbasis. So, we will form $d + 1$ such classes which will give us our $d + 1$ MUBs.

Let's first prove a useful lemma:

Lemma 1. *There are at most d pairwise orthogonal commuting unitary matrices in $\mathcal{B}(\mathcal{H})$.*

Proof. Let $A_1, \dots, A_m$ be pairwise orthogonal commuting unitary matrices in $\mathcal{B}(\mathcal{H})$. Then they can be simultaneously diagonalized, so there exists a unitary matrix S such that

$$B_j = S A_j S^\dagger,$$

where B_j is diagonal, for all j .

$$\langle B_j, B_k \rangle = \text{Tr}(B_j^\dagger B_k) = \text{Tr}(S A_j^\dagger S^\dagger S A_k S^\dagger) = \text{Tr}(S A_j^\dagger A_k S^\dagger) = \text{Tr}(A_j^\dagger A_k) = \langle A_j, A_k \rangle$$

Since A_j, A_k are orthogonal,

$$\langle B_j, B_k \rangle = \delta_{jk}$$

Let $|b_j\rangle \in \mathbb{C}^d$ denote the diagonal entries of B_j . Then

$$\langle b_j, b_k \rangle = \text{Tr}(B_j^\dagger B_k) = \delta_{jk}.$$

Hence, the vectors

$$|b_1\rangle \dots |b_m\rangle$$

are mutually orthogonal in $\mathbb{C}^d$. Therefore, there can be at most d such vectors, so $m \leq d$. $\square$

Theorem 3. *If there is a maximal commuting class of orthogonal unitary matrices in $\mathcal{B}(\mathcal{H})$, then there is a set of $d + 1$ MUB.*

Proof. Let $B = \{U_1, U_2, \dots, U_{d^2}\}$ be a maximal orthogonal unitary basis for $\mathcal{B}(\mathbb{C}^d)$. We say that B is a maximal commuting basis of orthogonal unitary matrices if it can be partitioned into,

$$B = \{I_d\} \cup C_1 \cup C_2 \cup \dots \cup C_{d+1},$$

where each class C_j contains exactly $d - 1$ commuting unitary matrices from B . Note, $\{I_d\} \cup C_j$ is a set of d commuting unitary matrices which we just proved is maximal.

Now,

$$C_j = \{U_{j,1}, \dots, U_{j,d-1}\},$$

and $U_{j,0} = I_d$

$\bar{C}_j = \{U_{j,0}, U_{j,1}, \dots, U_{j,d-1}\}$ forms a maximal commuting set.

Define

$$\mathcal{T}_j = \{|e_1^{(j)}\rangle, \dots, |e_d^{(j)}\rangle\},$$

$1 \leq j \leq d+1$ which are eigenvectors of $\bar{C}_j$. Elements of $\bar{C}_j$ commute among themselves, so they have common eigenbases.

for $0 \leq t \leq d-1$, diagonalized matrices of class $\bar{C}_j$ can be written as

$$D_{j,t} = \sum_{k=1}^d \lambda_{j,t,k} |e_k^j\rangle \langle e_k^j|$$

Define $M_j = \begin{pmatrix} \lambda_{j,0,1} & \lambda_{j,0,2} & \cdots & \lambda_{j,0,d} \\ \lambda_{j,1,1} & \lambda_{j,1,2} & \cdots & \lambda_{j,1,d} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{j,d-1,1} & \lambda_{j,d-1,2} & \cdots & \lambda_{j,d-1,d} \end{pmatrix}$

First row of this matrix is just $(1, 1, \dots, 1)$ from Identity Matrix.

$$\text{Tr}(U_{1,t_1}^\dagger U_{2,t_2}) = d \delta_{t_1 0} \delta_{t_2 0} \implies \langle U_{1,t_1}, U_{2,t_2} \rangle = 0.$$

Actually,

$$\text{Tr}(U_{1,t_1}^\dagger U_{2,t_2}) = \sum_{k_1, k_2=1}^d \lambda_{1,t_1,k_1}^* \lambda_{2,t_2,k_2} |\langle e_{k_1}^1 | e_{k_2}^2 \rangle|^2 = d \delta_{t_1,0} \delta_{t_2,0}, 0 \leq t_1, t_2 \leq d-1$$

This can be written as $AP = \Lambda$ where,

$$\Lambda = \begin{bmatrix} d \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad P = \begin{bmatrix} |\langle e_1^1 | e_1^2 \rangle|^2 \\ \vdots \\ |\langle e_d^1 | e_d^2 \rangle|^2 \end{bmatrix}$$

$$A = M_1^* \otimes M_2, \quad \text{then} \quad |\langle e_{t_1}^1 | e_{t_2}^2 \rangle|^2 = \frac{1}{d}, \quad 0 \leq t_1, t_2 \leq d-1.$$

By repeating the same argument for the classes C_j and C_k , we conclude that $\{\mathcal{T}_1, \dots, \mathcal{T}_{d+1}\}$ is a set of MUB. $\square$

References

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